

FLCA: A Flexible Coding Approach Through 5G NR LDPC Codes

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Abstract—This letter proposes a flexible coding approach (FLCA) for quasi-cyclic low-density parity-check (QC-LDPC) codes based on the base graphs and lifting factors in 5G new radio (NR). The FLCA can achieve almost all code rates and blocklengths supported by 5G physical layer using a novel shortening method which is different from those used in 5G NR. In the proposed FLCA, the shortening patterns are optimized according to the code rates and modulation schemes, including quadrature phase shift keying, 16-ary and 64-ary quadrature amplitude modulations. Our proposed method is fully applicable to the existing LDPC codes of 5G and makes the encoding more flexible. Both the numerical thresholds of PEXIT analysis and the simulation results demonstrate that the proposed coding approach can bring noticeable performance gain over the current LDPC codes used in 5G NR.

Index Terms—5G new radio, QC-LDPC codes, lifting factor, multi-lengths.

I. INTRODUCTION

LOW-DENSITY parity-check (LDPC) codes first proposed by Gallager in his doctoral thesis in 1962 [1], almost were not attracted any attention due to limited hardware support at that time until they were rediscovered by MacKay [2]. Since the 1990s, LDPC codes have been increasingly focused on due to their excellent error-correction performance and high-throughput iterative decoding architecture [3] and have been widely used in various communication systems, such as WiMAX, DVB and ATSC [4]. Nowadays, LDPC codes have been recommended by 3GPP as the channel coding scheme for the enhanced mobile broadband data channel of 5G new radio (NR) systems, which are designed to flexibly support the incremental redundancy hybrid automatic repeat request as well as a wide range of blocklengths and code rates.

Puncturing, shortening, and lifting are the common ways to make the transmission rates and lengths of quasi-cyclic (QC)-LDPC codes more flexible. Many contributions [5] have dealt with constructing rate-compatible LDPC codes which

are capable of realizing multi-rate codes using only a so-called mother code. Reference [6] proposes a rate-compatible scheme that combines the shortening and puncturing of information bits and parity-check bits, respectively. In [7], a random puncturing analysis of the LDPC code ensembles is presented which provides an efficient way to predict the thresholds. Several methods proposed in [8], [9], and [10] are employed in the constructions of QC-LDPC codes, which are especially suitable for parallel decoding implementations due to their QC structures consisting of cyclic shifted identity sub-matrices.

To optimize the LDPC codes for a specific transmission scenario, the researchers have designed various analytical tools. Some numerical analysis of using protograph extrinsic information transfer (PEXIT) [11], [12] have been developed to predict the decoding thresholds for protograph-based LDPC codes in different applications. Moreover, the PEXIT also can be used to evaluate the decoding performance of QC-LDPC codes with punctured and shortened bits, where the puncturing is inherent in the PEXIT analysis and the shortening can be easily performed by column erasure of the protograph.

Furthermore, the QC-LDPC codes can directly support the coding scheme with multiple lengths by using different lifting factors together with a base matrix. The 5G NR standard [13] offers two base graphs (BG1 and BG2), where BG1 that we mainly focus on supports the maximum block length 8448 bits and the range of code rates $\frac{1}{3} \sim \frac{11}{12}$. The positions and values of shift values are given by the protocol for each base matrix and the number of columns in BG1 ranging from 26 to 68 is variable according to the practical code rate. However, in order to meet the requirements of a wide range of blocklengths, the current coding scheme of 5G NR contains total 51 lifting factors in eight sets, which are too large to take the code design with each block length into careful consideration. Therefore, reducing the number of lifting factors as much as possible can make the optimization of 5G NR codes with multi-lengths more efficient.

In this letter, we propose a flexible coding approach (FLCA) which adopts a novel shortening method and changes the lifting construction of 5G LDPC codes. Unlike the rate-matching method in the 5G standard, the FLCA introduces more shortened bits by choosing a slightly larger lifting factor. The corresponding shortening patterns are optimized by the PEXIT analysis for various code rates and modulation schemes including quadrature phase shift keying (QPSK) and 2^q -ary quadrature amplitude modulation (QAM). Only seven sets of lifting factors are required for FLCA to achieve almost all blocklengths supported by 5G NR, while eight sets of lifting factors are used in the 5G standard. Even though fewer lifting factors are used, the simulation results demonstrate that our proposed scheme can achieve significant performance gain over the current LDPC coding scheme in 5G NR.

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II. PRELIMINARIES

An $M \times N$ parity-check matrix $\mathbf{H}$ of a QC-LDPC code is constructed from an $M_b \times N_b$ base matrix $\mathbf{H}_b$ by a positive integer Z known as the lifting factor, where $M = M_b \times Z$, $N = N_b \times Z$. Let $K = N - M$ be the code dimension (assuming $\mathbf{H}$ is of full rank) and $K_b = N_b - M_b$ be the number of columns corresponding to the information bits in the base matrix. The so-called *shift value* $P_{i,j} \in \{-1, 0, \dots, Z-1\}$, $0 \leq i \leq M_b - 1, 0 \leq j \leq N_b - 1$, of the base matrix will be replaced by an all-zero matrix or a circulant permutation matrix (CPM) of size Z when lifting the base matrix to a parity-check matrix. The CPM corresponding to an entry $P_{i,j} \geq 0$, is the identity matrix $\mathbf{I}$ cyclic right-shifted by $P_{i,j}$ times while an all-zero matrix is mapping from $P_{i,j} = -1$.

Two base graphs (BG1 and BG2) used in 5G NR both employ the raptor-like structure composed of five parts, denoted by **A**, **B**, **C**, **D** and **E**, respectively. A 5G LDPC code is lifted from BG1 or BG2 by a lifting factor Z and the shift values. All lifting factors supported by 5G LDPC codes are of the form $Z = x \times 2^j$ ranging from 2 to 384 with $x \in \{2, 3, 5, 7, 9, 11, 13, 15\}$ and $j = 0, 1, \dots, 7$. According to different x , a total of 51 lifting factors are placed into eight sets, and each factor is associated with the specified shift values $P_{i,j}$. Due to the QC structure, a codeword $\mathbf{C} = (\mathbf{c}_0, \mathbf{c}_1, \dots, \mathbf{c}_{N_b-1})$ can be written as N_b sub-vectors, each of which corresponds to a column in the base matrix and contains Z coded bits.

Example 1: A rate- $\frac{11}{15}$ base matrix¹ from the BG1 is illustrated in Fig. 1. This base matrix has a 4×26 core matrix, constituted by **A** and **B**, and supports the highest code rate of $\frac{11}{12}$. The rate extension matrix, formed by **D**, an all-zero matrix **C** and an identity matrix **E**, has a rate-compatible structure that can achieve the lowest code rate $\frac{1}{3}$. Additionally, a codeword of the sub-vector form is also shown in Fig. 1, which corresponds to the rate- $\frac{11}{15}$ base matrix. The information part with $K_b = 22$ sub-vectors is marked in blue, while the remaining $M_b = 10$ sub-vectors corresponding to parity-check columns are shown in green. Note that the first two sub-vectors $\mathbf{c}_0$ and $\mathbf{c}_1$ in the information part are always punctured, which are indicated by shades of blue.

Puncturing and shortening are two common techniques to adjust the blocklength and code rate for a given code. Although these two techniques are widely used, we would like to give their definitions for clearer representation of our proposed coding scheme.

Definition 1 (shorten): A bit b is said to be shortened if it is a known information bit not transmitted over the channel and its state $b \in \{0, 1\}$ is completely known for both the encoder and decoder.

Definition 2 (puncture): A bit b is said to be punctured if it is an unknown information bit or parity bit not transmitted over the channel and the decoder has no information about b , i.e., decoder takes $\Pr\{b = 0\} = \Pr\{b = 1\} = \frac{1}{2}$.

¹We note that in the technical specification involving 5G channel coding [13], encoding is based on the whole BG1 (rate- $\frac{1}{3}$) and then many bits will be punctured at the end of the coded sequence to achieve high rates. It is equivalent to taking a proper submatrix from the whole BG1 for encoding because **E** is an identity matrix. In this letter, the base matrix of size $M_b \times N_b$ corresponds to such a submatrix.

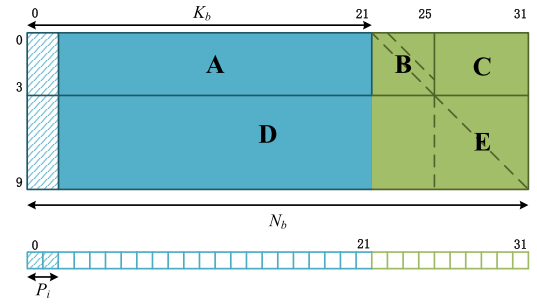


Fig. 1. The structures of the base matrix and the corresponding codeword for 5G NR LDPC codes with $R = \frac{11}{15}$.

From the above definitions, we know that both puncturing and shortening reduce the actual code length for transmission. But the shortening also reduces the code dimension because such shortened bits are known for the decoder. For an (N, K) code, if we puncture P bits and shorten S bits, the actual code rate will be

$$R = \frac{K - S}{N - P - S}. \quad (1)$$

Clearly, we can define $K_{\text{ef}} := K - S$ to be the *effective dimension* and $N_{\text{ef}} := N - P - S$ to be the *effective code length* for transmission. For a soft decoder using bit log-likelihood ratios (LLRs), the LLR of a shortened bit can be sufficiently large while the LLR of a punctured bit can only be set to zero. In this letter, we fix the states of shortened bits to be zeros.

The 5G LDPC codes can achieve a large range of code lengths and rates by puncturing and shortening.² When K_{ef} and R are given, a proper Z needs to be selected first and then the settings of puncturing and shortening can be determined according to the technical specification [13]. In 5G NR, except for the first two columns that are required to be punctured, the other punctured bits are distributed at the end of the parity-check bits. The positions of the shortened bits are placed at the end of the information bits. Denote by $\mathcal{Z}_0 = \{Z_1, \dots, Z_{51}\}$ the ordered set of all lifting factors supported by the standard. We can form a candidate set of lifting factors as $\mathcal{Z} = \{Z \in \mathcal{Z}_0 | Z \geq \frac{K_{\text{ef}}}{K_b}\}$. The lifting factors in $\mathcal{Z}$ guarantee that the code has large enough dimension to fill the information bits, and the redundant dimension will be reduced by shortening. In 5G standard, the lifting factor is uniquely determined by the criterion

$$Z_a = \min_{Z \in \mathcal{Z}} \{Z\}. \quad (2)$$

In the next section, we propose a flexible coding approach (FLCA) of QC-LDPC codes, which gives a different criterion for choosing the lifting factor as well as the settings of shortening and puncturing.

III. FLCA WITH OPTIMIZED SHORTENING PATTERNS

A. The Procedure of FLCA

Recall that a codeword is represented by the sub-vector form which is consistent with the QC structure. The proposed FLCA

²The shortened bits are referred to as “filler bits” in the technical specification [13].

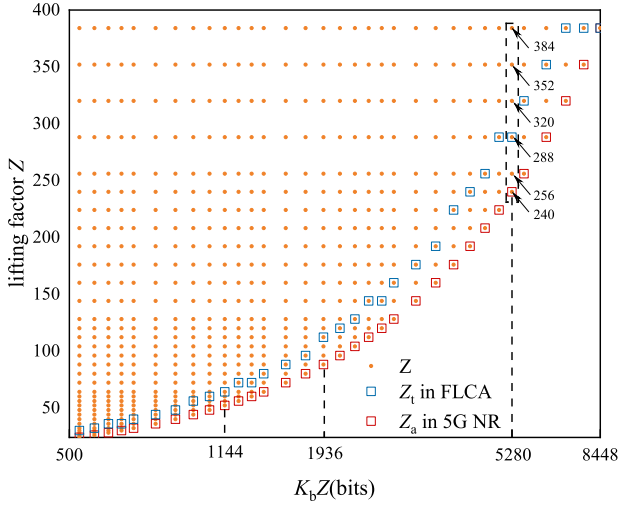


Fig. 2. The lifting factors versus different blocklengths for BG1.

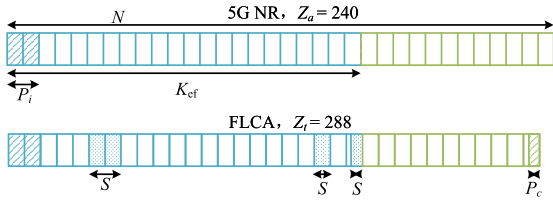


Fig. 3. The code structures of 5G NR and FLCA with $K_{\text{eff}} = 5280$ and $R = \frac{11}{16}$.

takes the following three steps to fulfill a given requirement (K_{eff}, R) : 1) Select a lifting factor Z_t ; 2) Determine the number of shortened bits S and the corresponding positions; 3) Determine the number of punctured bits P . The first two steps follow a different criterion from that used in 5G NR.

Firstly, we give a criterion to select a lifting factor from the candidate set of 5G which is equal to or slightly larger than Z_a . To avoid noticeable extra complexity caused by introducing too many shortened bits, $m \in \{2, 3, 4, 5\}$ sub-vectors of a codeword are allowed to be shortened. Define a new candidate set $\tilde{\mathcal{Z}} := \mathcal{Z} \cap [\frac{K_{\text{eff}}}{K_b - 2}, \frac{K_{\text{eff}}}{K_b - 5}]$. The criterion for the first step is

$$Z_t = \begin{cases} \max_{Z \in \tilde{\mathcal{Z}}} \{Z\}, & 500 \leq K_{\text{eff}} \leq 7680, \\ 384, & 7680 < K_{\text{eff}} \leq 8448. \end{cases} \quad (3)$$

Following this criterion, the difference of Z_a (5G standard) and Z_t (FLCA) is shown in Fig. 2. The dots in the figure are the elements in the candidate set $\mathcal{Z}$, and blue and red boxes correspond to Z_t and Z_a , respectively.

For the second step, we determine the shortening and puncturing positions. Calculate the length of shortened information bits $S = K_b Z_t - K_{\text{eff}}$. In FLCA, the positions of the shortened bits are distributed according to the specifically optimized patterns for different lengths. Once S is determined, define a parameter $m := \lfloor \frac{S}{Z} \rfloor$, where $\lfloor \cdot \rfloor$ is the flooring function. The shortened positions chosen by FLCA cover m whole sub-vectors and $S - mZ$ positions in some sub-vector if $S - mZ > 0$. Clearly, $S - mZ < Z$. A shortening pattern is denoted as an index set $\mathcal{S}_m = \{s_1, \dots, s_m\} \forall s_i \in [2, K_b - 1], i = 1, \dots, m$. We will shorten all sub-vectors c_i such that $i \in \mathcal{S}_m$. After that, the remaining $S - mZ$ shortened

bits will be placed in the last bit positions of the sub-vector c_{i^*} , where i^* is the maximum integer in $\{2, 3, \dots, K_b - 1\} \setminus \mathcal{S}_m$.

Thirdly, we need to choose a proper submatrix from BG1, i.e., determine N_b , and then puncture some bits to fulfill the rate requirement R . The positions of punctured bits in FLCA are same as those in 5G NR. Recall $R = \frac{K_{\text{eff}}}{N_{\text{eff}}}$, and (K_{eff}, R) is given as the requirement. We denote the number of punctured bits in the first two columns, i.e., c_0, c_1 , by $P_i = 2Z$. Hence, the column number of the base matrix N_b can be obtained by $N_b = \lceil \frac{N_{\text{eff}} + S + P_i}{Z} \rceil$, where $\lceil \cdot \rceil$ denotes the ceiling function. Once the entire base matrix is achieved, the whole parity-check matrix $\mathbf{H}$ can be uniquely lifted from the base matrix according to [13]. In this way, $\mathbf{H}$ has $N_b Z_t$ columns, so we can obtain $P = N_b Z_t - N_{\text{eff}} - S$. Finally, we need to set $P_c = P - P_i$ parity-check bits at the end to be punctured. One can check that the rate requirement is fulfilled because $\frac{K_b Z_t - S}{N_b Z_t - P - S} = \frac{K_{\text{eff}}}{N_{\text{eff}}} = R$.

After puncturing and shortening, only N_{eff} coded bits are sent to the interleaver and the interleaved sequenced will be mapped into QAM symbols. Please refer to [13] and [14] for the details of interleaving and modulation. We end this section with the following example.

Example 2: Assume that we desire to transmit $K_{\text{eff}} = 5280$ bits at the code rate $R = \frac{11}{16}$. Thus, the effective code length is $N_{\text{eff}} = \lceil \frac{K_{\text{eff}}}{R} \rceil = 7680$. Hence, the lifting factor should be no less than $\frac{K_{\text{eff}}}{K_b} = 240$ and so the candidate set is formed as $\mathcal{Z} = \{240, 256, 288, 320, 352, 384\}$, as shown in Fig. 2. Using the criterion (2) for 5G NR, the selected lifting factor is $Z_a = 240$, and the number of shortened bits is $S = 0$. We then obtain all parameter: $P_i = 480$, $N_b = 34$, and $P_c = 0$. While, FLCA takes $Z_t = 288$ according to (3) and has $S = 1056$ shortened bits. Similarly, we can obtain the other parameters: $P_i = 576$, $N_b = 33$, and $P_c = 192$. For the shortening pattern, we have $m = \lfloor \frac{1056}{288} \rfloor = 3$. Assuming 64QAM is applied, the shortening pattern is set to be $\mathcal{S}_3 = \{5, 6, 19\}$ according to Table I, which is optimized by the method presented in the next subsection.

The structures of two codewords corresponding to 5G NR and FLCA are shown in Fig. 3. The blue and green blocks represent the information and parity-check sub-vectors, respectively. The shaded area indicates the columns of punctured bits, while the dotted area represents the positions of shortened bits. In this case, the codeword for 5G NR has one more sub-vector, corresponding to one column in the base matrix, than FLCA. On the other hand, FLCA has a larger lifting factor Z_t compared with Z_a for 5G NR.

B. The Optimized Shortening Patterns

The PEXIT analysis can be used to compute the decoding threshold of an LDPC-coded modulation system. We utilize the PEXIT analysis to calculate decoding thresholds for various shortening patterns and then some optimized shortening patterns can be obtained. Due to the space limit, we only show how to initialize the mutual information (MI) for N_b VNs, which is affected by the puncturing, shortening, and interleaving schemes. The remaining steps of the PEXIT analysis are the same as the PEXIT analysis for BPSK systems [11].

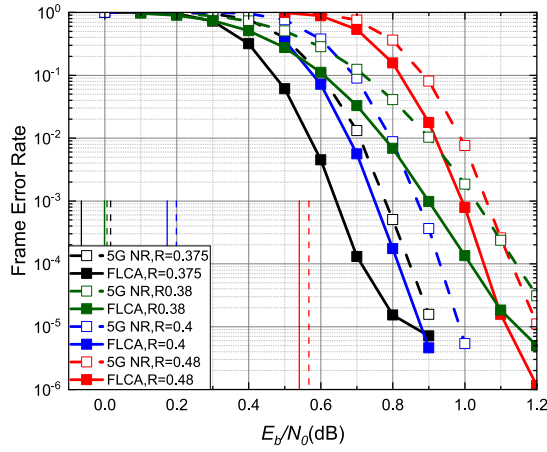


Fig. 4. FER performance of 5G NR codes and the optimized codes with QPSK.

For 2^q QAM, let B_i , $i \in \{0, \dots, q-1\}$, and L_i , $i \in \{0, \dots, q-1\}$, be the random variables of the i -th bit position of a QAM symbol and its corresponding LLR, respectively. The MI between B_i and L_i , denoted by $I(B_i; L_i)$, can be obtained by theoretical computation or Monte Carlo simulation. Each VN i in the base matrix corresponds to Z coded bits $\mathbf{c}_i = (c_{i,0}, \dots, c_{i,Z-1})$, and each coded bit has $q+2$ states, be mapped to the j -th, $j = 0, 1, \dots, q-1$, bit position, be punctured, or be shortened. Fixing a modulation order, an interleaving scheme, and a shortening pattern, we can know for each VN i , Λ_i^j , Λ_i^P , and Λ_i^S bits (for VN i) are mapped to the j -th bit position, punctured, and shortened, respectively. Therefore, for each VN i , the initial channel MI I_{ch}^i is obtained by

$$I_{ch}^i = 0 \times \lambda_i^P + 1 \times \lambda_i^S + \sum_{j=0}^{q-1} \lambda_i^j I(B_j; L_j) \quad (4)$$

where $\lambda_i^P = \frac{\Lambda_i^P}{Z}$, $\lambda_i^S = \frac{\Lambda_i^S}{Z}$, and $\lambda_i^j = \frac{\Lambda_i^j}{Z}$. Note that $\lambda_i^P + \lambda_i^S + \sum_{j=0}^{q-1} \lambda_i^j = 1$. The remaining steps of PEXIT analysis are the same as those in [11].

In the FLCA scheme, the constraints $2Z \leq S \leq 5Z$ and $m \in \{2, 3, 4, 5\}$ are implied due to (3). Therefore, we use an exhaustive search to find the optimal shortening pattern, which takes $\binom{K_b-2}{m}$ times of PEXIT computations for each code. The code rate supported by FLCA ranges from $\frac{1}{3}$ to $\frac{11}{13}$, and the optimized shortening patterns S_m for various coded modulation schemes are shown in Table I.

IV. SIMULATION RESULTS AND ANALYSIS

We evaluate the performance of QC-LDPC codes obtained by the proposed FLCA and 5G NR with different blocklengths and code rates, respectively. Additive white gaussian noise channels and the layered normalized min-sum (LNMS) decoding algorithm with 30 iterations are assumed.

Table II lists the parameters of coded modulation schemes used in simulations, where the QC-LDPC codes are all selected from BG1 of 5G NR. Thus, we can see that the spectral efficiency η gradually increases with QPSK, 16QAM and 64QAM. The numerical thresholds (T_h) of these schemes analyzed by PEXIT are shown in the last row of Table II.

TABLE I
THE OPTIMIZED SHORTENING PATTERNS S_m FOR
VARIOUS CODED MODULATION SCHEMES

q	R	S_2	S_3	S_4	S_5
2	$\frac{1}{3} \sim \frac{11}{19}$	11, 21	4, 11, 12	2, 3, 13, 16	3, 4, 10, 15, 16
4	$\frac{11}{30} \sim \frac{11}{19}$	6, 15	9, 15, 20	5, 9, 16, 17	3, 5, 9, 15, 19
	$\frac{11}{19} \sim \frac{11}{17}$	12, 16	10, 19, 21	6, 16, 19, 21	10, 12, 15, 19, 21
6	$\frac{11}{24} \sim \frac{11}{13}$	6, 15	5, 6, 19	5, 9, 16, 17	6, 11, 15, 16, 19

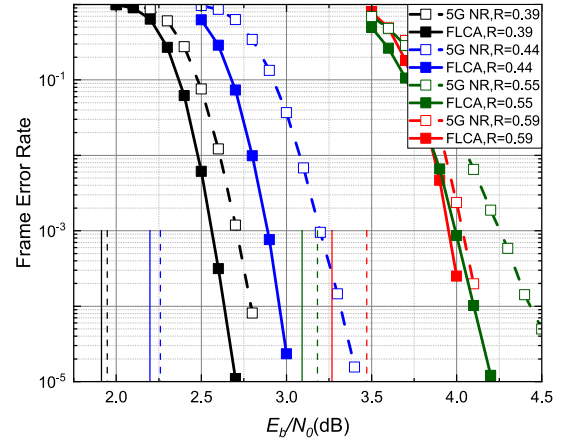


Fig. 5. FER performance of 5G NR codes and the optimized codes with 16QAM.

It can be easily seen that the QC-LDPC codes constructed by FLCA we proposed have better thresholds compared to the coding schemes in 5G NR. Next, the advantages of FLCA will be illustrated through frame error rate (FER) performance of these codes with different modulations.

In all figures of FER performance comparisons, four colors (red, blue, green and black) of FER curves correspond to a coded modulation scheme with different blocklengths and code rates. The dashed lines with hollow patterns represent the FER performance of the original codes used in 5G with the legend “5G NR”, which corresponds to “5G” in Table II. The solid lines with solid squares denote those codes constructed by our proposed scheme with the legend “FLCA”. The vertical lines without symbols are the numerical thresholds of these coded modulation schemes.

In Fig. 4, the codes with $K_{ef} = 6912$, $K_{ef} = 2432$ and $K_{ef} = 7680$ have been chosen and their rates are 0.375, 0.38, 0.4, and 0.48, respectively. It is obvious that there is a noticeable gap between the thresholds of the base matrices used by the protocol and those obtained by FLCA. Fig. 4 demonstrates that all of the codes in FLCA perform better than those in 5G standard and achieve about 0.1 dB gain at an FER of 10^{-4} . It is noteworthy that the codes with rates 0.375 and 0.4 share the common mother code whose number of columns is 54 and the lifting factor is 384, which indicates that the FLCA is more flexible in terms of code rate adjustment.

The shortened bit positions S_m of the QC-LDPC codes in FLCA are determined according to Table I. We select the base matrices of 5G NR LDPC codes with $R = 0.39, 0.44, 0.55$ and 0.59 in the coded modulation scheme with 16QAM. Fig. 5 shows the performance comparisons of the FLCA and 5G NR

TABLE II
THE PARAMETERS AND THRESHOLDS OF DIFFERENT CODED MODULATION SCHEMES WITH INCREASING SPECIAL EFFICIENCY η

	QPSK								16QAM								64QAM							
Method	5G	FLCA	5G	FLCA	5G	FLCA	5G	FLCA	5G	FLCA	5G	FLCA	5G	FLCA	5G	FLCA	5G	FLCA	5G	FLCA	5G	FLCA	5G	FLCA
η	0.75		0.76		0.8		0.96		1.56		1.76		2.2		2.36		2.88		3.3		4.2		4.98	
R	0.375		0.38		0.4		0.48		0.39		0.44		0.55		0.59		0.48		0.55		0.7		0.83	
K_{ef}	6912		2432		7680				7296		6528		3808		6528		7680		4500		7296		1152	
Z_a/Z_t	320	384	112	128	352	384	352	384	352	384	320	384	176	224	320	384	352	384	208	256	352	384	56	64
N_b	60	54	60	55	57	54	48	46	57	54	51	46	43	39	39	36	48	46	42	39	33	32	29	0.28
S	128	1536	32	384	64	768	64	768	448	1152	512	1920	64	1120	512	1920	64	768	76	1132	448	1152	80	256
P_c	0	0	64	0	96	0	0	0	96	0	192	0	172	188	192	0	0	0	60	156	96	0	52	28
$T_h(\text{dB})$	0.02	-0.07	0.006	-0.001	0.2	0.17	0.57	0.54	1.95	1.91	2.26	2.2	3.18	3.09	3.47	3.27	4.89	4.8	5.68	5.58	7.51	7.49	9.19	9.12

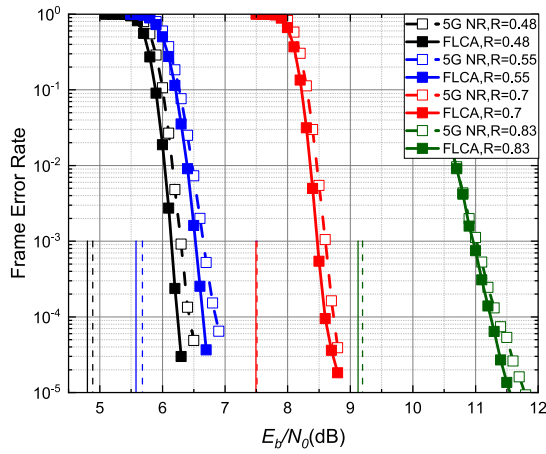


Fig. 6. FER performance of 5G NR codes and the optimized codes with 64QAM.

coded schemes. As we can see, the proposed codes based on BG1 demonstrate significant improvements in FER simulation results and numerical thresholds, where the coded modulation with $K_{\text{ef}} = 3808$ and rate 0.55 achieves a gain of 0.35 dB at an FER of 10^{-4} .

In Fig. 6, four groups of QC-LDPC codes with different blocklengths, $K_{\text{ef}} = 7680$, $K_{\text{ef}} = 4500$, $K_{\text{ef}} = 7296$ and $K_{\text{ef}} = 1152$, are compared. With the limited number of iterations, these coded modulation schemes optimized by the proposed FLCA obviously outperform in comparison to those in the protocol, where our codes achieve a gain of approximately 0.2 dB at an FER of 10^{-4} . The simulation results of FER are consistent with the numerical thresholds of PEXIT analysis in Table II, which shows the superiority of the proposed FLCA.

V. CONCLUSION

We propose a flexible coding approach called FLCA for 5G NR LDPC codes based on BG1, which can be evaluated by the PEXIT analysis with limited iterations. The proposed approach selects a slightly larger value Z_t and optimized

shortening patterns to perform encoding, which results in more shortened bits and more flexibility for multiple lengths and rates adjustment. Fewer lifting factors are used in the FLCA than 5G NR. The FLCA covers almost all the code rates and code lengths supported by the 5G protocol. Simulation results show that our proposed method can achieve noticeable performance gains over 5G NR.

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